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```
% Problem 3a - Bode Plot
```

```
s = tf('s');
```

```
% Define each block
```

```
Kc = 1; % Set to 1 initially; adjust as needed for part (b)
```

```
Gc = Kc * (2*s + 1)/(0.1*s + 1);
```

```
Gp = 0.4/(s*(5*s + 1));
```

```
Gv = 2/(0.2*s + 1);
```

```
Gm = 1;
```

```
% Open-loop transfer function
```

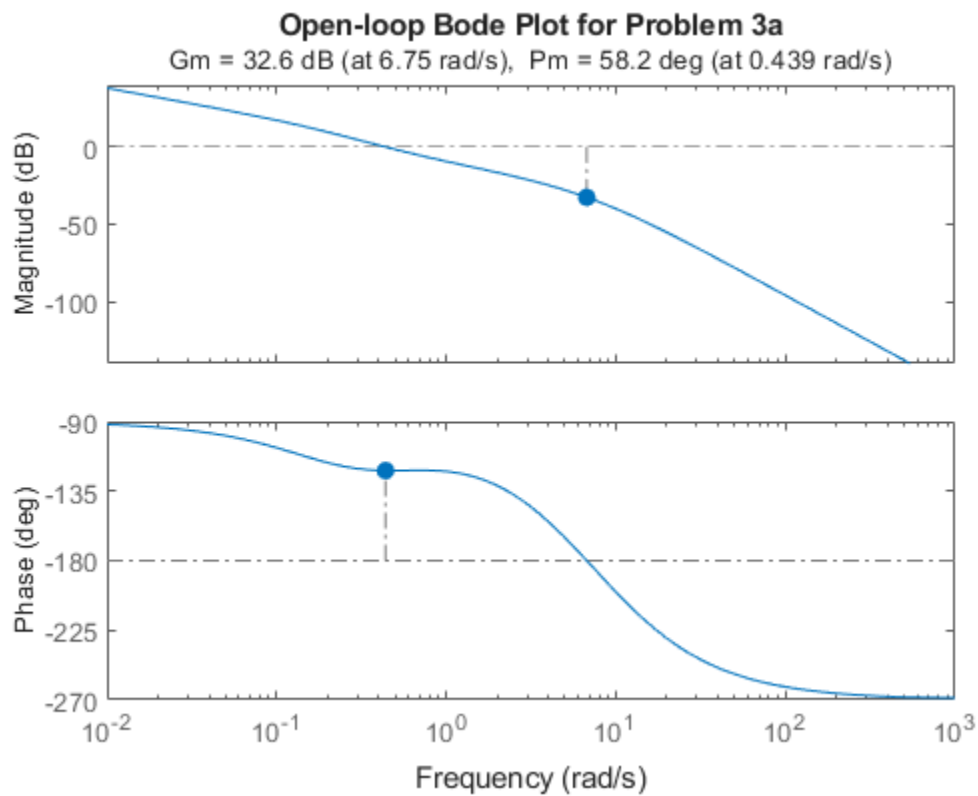
```
Gol = Gc * Gp * Gv * Gm;
```

```
% Bode plot
```

```
figure;
```

```
margin(Gol); % shows Bode plot with gain and phase margins
```

```
title('Open-loop Bode Plot for Problem 3a');
```



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